When Field-Wise Metrics Miss Hierarchical Consistency: A Controlled Study on the AI CUP ESG Promise Verification Dataset

Sheng-Xun Chen
National Central University
Taipei, Taiwan
tom1030507@gmail.com

Chun-Wei Chen
National Central University
Taipei, Taiwan
aaronwayway@gmail.com

Hsin-Chen Han
National Central University
Taipei, Taiwan
0720samson@gmail.com

Shi-Yu Huang
National Central University
Taipei, Taiwan
huangshiyu0318@gmail.com

ABSTRACT

ESG promise verification jointly predicts promise status, verification timeline, evidence status, and evidence quality under hard parent–child constraints. We compare seven decision rules on identical cross-fitted base probabilities and test rows from the AI CUP VeriPromiseESG development set. Independent argmax (M0) emits invalid tuples on 12.6% of document-disjoint predictions, whereas projection and 17-state decoding are legal by construction. For projection versus independent argmax (M1–M0), the official field-wise weighted macro-F1 shows no detectable difference: effect −0.001; 95% CI $[-0.006, 0.003]$; $p_{\text{Holm}} = 1.000$. Post-hoc path-constrained weighted macro-F1, changing only credit for children unsupported by predicted ancestors, gives effect +0.004 and CI $[0.001, 0.007]$. Its $p_{\text{Holm}} = .025$. Pre-specified tuple accuracy gives effect +0.035 and CI $[0.028, 0.043]$; $p_{\text{Holm}} = .001$. Conversely, uncalibrated 17-state decoding versus uncalibrated projection (M4–M1) gives a tuple accuracy effect of −0.006 and CI $[-0.010, -0.002]$. Its $p_{\text{Holm}} = .028$. All intervals are uncorrected and do not determine decisions. On this dataset and Chinese backbone, field-wise scoring can understate the utility of validity-preserving decisions.

KEYWORDS

ESG, promise verification, hierarchical classification, constrained decoding, evaluation metrics

1 INTRODUCTION

ESG promise verification is not four unrelated classification problems. The predicted promise status, verification timeline, evidence status, and evidence quality describe one claim, and their parent–child implications determine whether the combined output is internally usable. For example, a paragraph predicted to contain no promise cannot coherently receive a substantive timeline, and a prediction of no evidence cannot coherently receive an evidence quality. Individually plausible field labels can therefore form an illegal tuple.

The official weighted macro-F1 nevertheless scores each field separately. It can award credit to a substantive child whose predicted ancestors do not support that child, while a rule that preserves validity must repair or replace it. This creates a gap between what the task structure requires and what the ranking metric rewards. We ask whether the official field-wise score reflects the value of enforcing the supplied hierarchy.

We study that question as a controlled decision comparison, not as a model or competition-performance claim. Seven rules consume identical cross-fitted base probabilities and test rows. Independent argmax emits 12.6% illegal tuples in the primary unseen-report evaluation, whereas the structured rules are legal by construction. Projection versus independent argmax has no detectable difference on the official metric ($p_{\text{Holm}} = 1.000$). The post-hoc C-wF1 contrast, differing by a single scoring change, is positive after Holm correction ($p_{\text{Holm}} = .025$); pre-specified tuple accuracy corroborates its direction ($p_{\text{Holm}} = .001$). Post-hoc hF also resolves M1–M0 ($p_{\text{Holm}} = .017$), but changes the aggregation. These results are limited to this dataset, backbone, and set of decision rules.

Our contributions are:

(1) We formulate AI CUP ESG promise verification as a multi-task problem constrained to 17 legal tuples out of 120 possible field combinations and audit the development data, including its severe class and state imbalance.
(2) We compare seven decision rules in a controlled design that holds the cross-fitted base probabilities and evaluated rows identical.
(3) We quantify a task–metric mismatch by comparing official weighted macro-F1 with post-hoc C-wF1. Only the scoring of unsupported children changes.
(4) We report the adverse tuple result for uncalibrated M4 versus M1 and protocol sensitivity. Under same-document evaluation, the headline C-wF1 finding does not replicate. Here, $p_{\text{Holm}} = .739$. The 0.012–0.015 estimand gaps are not bias estimates.

2 RELATED WORK

The AI CUP task uses the same four-part formulation as PromiseEval: identify an ESG promise, determine whether it has actionable evidence, assess the clarity of the promise–evidence pair, and assign a verification horizon [1, 2]. This joint verification problem differs from ClimateBERT-NetZero, which detects corporate and national net-zero or reduction targets and supports analysis of their ambition, rather than jointly assessing promise status, evidence, evidence quality, and timing [8].

These dependent outputs form a compact instance of hierarchical text classification, in which labels occupy a supplied structure and coherent predictions respect its ancestor relations [7]. The AI CUP hierarchy comes from the task's official field logic: we neither discover the hierarchy nor propose a new hierarchical architecture. Instead, we compare decision rules over the same field-wise probabilities while preserving or ignoring that given structure.

Yu et al. introduced path-constrained MicroF1 and MacroF1, under which a correct node receives credit only when all its ancestors are also correct [9]. Ji et al. later named this family *C-metrics* [6]. Our C-wF1 adapts Yu et al.'s principle to this field-based task and its official field-weighted macro-F1; it is therefore not the original C-MacroF1.

Plaud et al. observed from point estimates that constrained F1 variants did not globally change method rankings in their experiments [7]. That observation and our post-hoc paired contrast inference answer different questions: ranking a collection of models is not the same estimand as a paired difference between decision rules on shared rows and probabilities. Moreover, our hF changes both consistency treatment and macro-versus-micro aggregation. Of the two post-hoc metrics, only C-wF1 isolates the treatment of ancestor-inconsistent predictions while retaining the official field aggregation.

3 TASK AND DATA

Outputs and legal states. The task predicts four fields for each paragraph. We study the AI CUP VeriPromiseESG development release; its label inventory and parent–child logic follow the official submission guide [1]. Promise status (PS) is *Yes* or *No*. Verification timeline (VT) is *already*, *within two years*, *between two and five years*, *more than five years*, or *N/A*. Evidence status (ES) is *Yes*, *No*, or *N/A*; evidence quality (EQ) is *Clear*, *Not Clear*, *Misleading*, or *N/A*. These fields obey hard implications: PS=No requires VT=ES=EQ=N/A, whereas PS=Yes requires one of the four substantive timelines and ES in {Yes, No}. In turn, ES=No requires EQ=N/A, whereas ES=Yes requires one of the three substantive quality labels. Thus the Cartesian label space has $2 \times 5 \times 3 \times 4 = 120$ combinations, but only

$$1 + 4(1 + 3) = 17 \tag{1}$$

legal tuples: one for PS=No, and for each of four timelines either one ES=No state or three ES=Yes quality states.

Development-set audit. The release contains 2,000 labeled development paragraphs from 49 source reports covering 50 companies, and it realizes 15 of the 17 legal states (Table 1). The rarest EQ label, *Misleading*, has only two instances; we therefore make no improvement or significance claim for that class.

Table 1: Dataset summary from the frozen study artifact.

Statistic	Development	Test
Paragraphs	2000	2000
duplicated across splits	1	1
Source reports (PDFs)	49	49
shared across splits	49	49
Companies	50	50
spanning >1 report	0	n/a
Legal states observed	15 / 17	n/a
Rarest classes		
within_2_years	34	n/a
Misleading	2	n/a

All 49 development reports also occur in the unlabeled competition test data, and one paragraph text occurs in both splits. This overlap describes sources and text only: because competition test labels are unavailable, we infer no label-derived test statistic. No company contributes more than one report; one report covers two companies. Consequently, in this corpus a document-disjoint split is necessarily company-disjoint, rather than a separate company-level experiment.

4 METHODS

4.1 Shared Base Model

We use `hfl/chinese-roberta-wwm-ext-large` as the fixed encoder. BERT-Large has 24 layers and hidden size 1024 [4]; the pinned HFL configuration matches those dimensions. Cui et al. document the checkpoint's Chinese RoBERTa-WWM family: it uses whole-word masking, extended pretraining data, and no next-sentence prediction [3]. We mean-pool its token representations and attach four independent linear heads, one per field. The encoder is shared, but neither it nor the heads condition on predicted ancestors. The input is paragraph text only; report URL, company, ticker, and page metadata are excluded.

The encoder and matching tokenizer are loaded from the pinned snapshot revision `a25cc9e05974bd9687e528edd516f2cfdb3f5db9`.

All fits use the same frozen fine-tuning recipe: maximum length 384, 12 epochs, minibatches of 8 with two-step gradient accumulation (effective size 16), and AdamW with backbone and head learning rates 2×10^{-5} and 10^{-4}. The schedule is cosine with 10% warm-up and layer-wise decay 0.9. The four cross-entropy losses use the official PS/VT/ES/EQ weights 0.20/0.15/0.30/0.35 and label smoothing 0.1. For Train size n, K_t classes present in field t, and class support $n_{t,c}$, the class weight is $\min\{10, n/(K_t\, n_{t,c})\}$ when $n_{t,c} \geq 5$, and 1 otherwise. Hierarchy masking is disabled. We average the last three epoch checkpoints and perform no label-driven early stopping.

4.2 Metric-Aware Decision Calibration

For field t and class c, decision calibration adds a class bias in log probability space:

$$z_{t,c}(x) = \log p_{t,c}(x) + b_{t,c}. \tag{2}$$

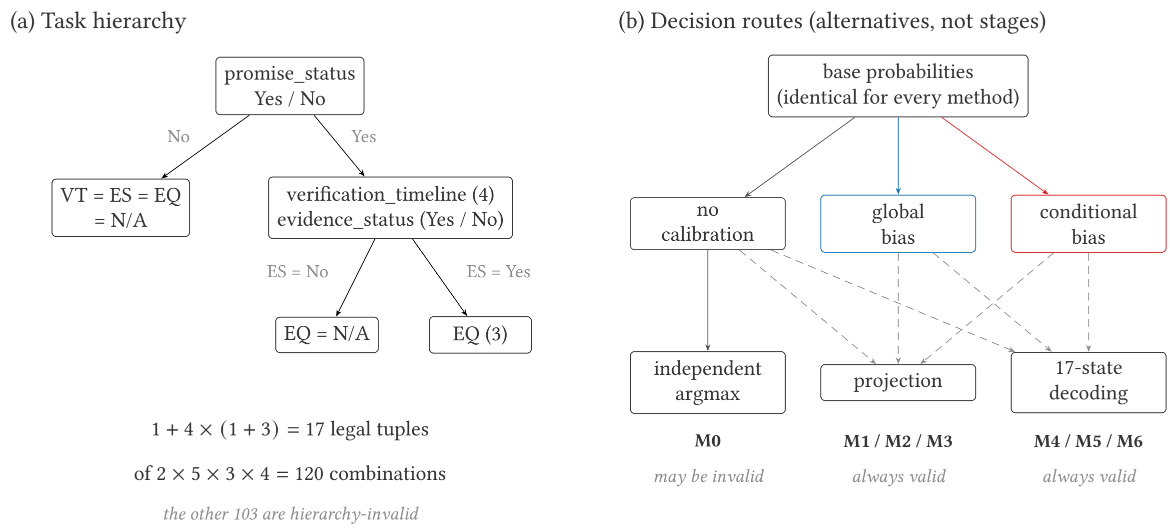

Figure 1: The task hierarchy and the alternative decision routes. All routes consume identical base probabilities; projection and 17-state decoding guarantee a legal tuple, whereas independent argmax does not.

This changes decision boundaries and is not a claim of probability calibration. Each field's bias vector is fitted by deterministic coordinate ascent with that field's macro-F1 on the held-out Calibration partition as the objective, before either structured output rule is applied. The four objectives therefore decouple. Within every rotation, global and conditional biases are fitted separately; that rotation's global fit is shared by M2/M5 and its conditional fit by M3/M6.

Global biases use every Calibration row. Conditional biases use all rows for PS, only gold PS=Yes rows for VT and ES, and only gold ES=Yes rows for EQ. The three child N/A classes are structurally absent from these subsets. Their biases are therefore unidentifiable and fixed as

$$b_{\mathrm{VT,N/A}} = b_{\mathrm{ES,N/A}} = b_{\mathrm{EQ,N/A}} = 0. \tag{3}$$

This has no effect on projection, which emits N/A only when its parent requires it. It does affect joint scores for (No, N/A, N/A, N/A); hence M6–M5 compares conditional with global estimation including their different estimable parameter sets.

Biases start at zero and are fitted by deterministic coordinate ascent over $[-3, 3]$ in increments of 0.05. Coordinates follow a fixed field and class order; a move requires a strict improvement, ties favor the smaller absolute bias (then the more negative value), and search stops after a pass without improvement or at 20 passes. Any estimable class absent from a Calibration partition retains bias zero.

4.3 Validity-Preserving Output Rules

Complete projection. Projection is greedy and top-down. At each field it selects the highest-scoring class allowed by the chosen ancestors. Under PS=No, it sets VT, ES, and EQ to N/A. Under PS=Yes, it excludes N/A from VT and ES and chooses each field's best substantive class. Under ES=No, EQ becomes N/A; under ES=Yes, projection chooses EQ's best substantive class. These rules completely enforce both directions of every parent implication. This bidirectional projection always returns one of the 17 legal states; a one-way N/A overwrite would not.

Joint valid-state decoding. The alternative scores every s in the legal set $\mathcal{S}$ and returns

$$\hat{y} = \arg\max_{s \in \mathcal{S}} \sum_t \alpha_t \{\log p_{t,s_t}(x) + b_{t,s_t}\}, \qquad \alpha_t = 1. \tag{4}$$

Fixing all task scales avoids giving the decoder additional tuned parameters.

Figure 1 summarizes seven *alternative* rules over identical cross-fitted base probabilities, not sequential stages: M0 is uncalibrated independent argmax; M1/M2/M3 use projection with respectively no, global, or conditional bias; and M4/M5/M6 use 17-state decoding with the same three bias choices. Only M0 can emit an illegal tuple.

5 EXPERIMENTAL DESIGN

We first define the shared evaluation protocols and inference. Table 2 then summarizes the controlled comparison interpreted in Section 6.

5.1 Cross-Fitted Evaluation Targets

We use five-way rotating three-way cross-fitting. For rotation k, Test is fold k, Calibration is fold $(k+1) \bmod 5$, and Train comprises the other three folds; the three partitions are mutually exclusive. After five rotations, every development paragraph has served as Test exactly once. We concatenate the 2,000 out-of-fold predictions and compute each metric once, rather than average fold-level F1 values whose present gold-label sets may differ. The three seeds 42, 123, and 456 each determine both a fold assignment and training randomness, so seed variation represents the complete pipeline.

The primary, document-disjoint protocol balances report row counts with a randomized greedy procedure. After a seeded shuffle,

Table 2: Main document-disjoint results from the frozen study artifact.

ID	Calibration	Decoding	Weighted F1	PS	VT	ES	EQ	Tuple Acc.	Invalid %
M0	None	Independent	0.572±0.003	0.796	0.464	0.650	0.424	0.359	12.6
M1	None	Projection	0.571±0.003	0.796	0.467	0.651	0.418	0.394	0.0
M2	Global	Projection	0.574±0.004	0.796	0.493	0.650	0.418	0.428	0.0
M3	Conditional	Projection	0.574±0.005	0.796	0.501	0.651	0.412	0.431	0.0
M4	None	17-state	0.571±0.005	0.799	0.468	0.648	0.420	0.388	0.0
M5	Global	17-state	0.576±0.001	0.796	0.493	0.649	0.423	0.430	0.0
M6	Conditional	17-state	0.567±0.009	0.784	0.494	0.636	0.414	0.426	0.0

it processes reports from largest to smallest and randomly places each in one of the two currently lightest folds. It estimates unseen-report generalization and, given the corpus audit, is also company-disjoint.

The same-document protocol instead jointly stratifies by gold PS, ES, and VT. It shuffles rows within each joint-label bucket and deals them round-robin to the five folds. Every report represented in Calibration or Test must also occur in Train through a different paragraph. This estimates seen-report, unseen-paragraph generalization. The protocols are distinct estimands, not biased and unbiased versions of one quantity.

Within each protocol and seed, M0–M6 consume the same Test rows and base probabilities.

5.2 Metrics and Inference

The primary metric is the official weighted macro-F1,

$$\text{wF1} = 0.20F_{\text{PS}} + 0.15F_{\text{VT}} + 0.30F_{\text{ES}} + 0.35F_{\text{EQ}}, \tag{5}$$

where each F_t is field-level macro-F1 over classes present in the gold labels being scored. Whole-tuple accuracy, which requires all four fields to match, was pre-specified. Path-constrained weighted macro-F1 (C-wF1) and hierarchical F1 (hF) were adopted post hoc. C-wF1 retains the official field weights but counts a substantive child prediction as wrong when its predicted ancestors do not support it. Following the hierarchical precision/recall/F1 set-overlap definition [7], our hF micro-averages over emitted non-N/A field–value nodes: hP is total gold–prediction overlap divided by predicted nodes, hR divides the same overlap by gold nodes, and hF is their harmonic mean. For an inconsistent emitted tuple, we do not add missing ancestors before scoring; legal M1–M6 are unaffected because the conventions coincide for them. hF changes both consistency treatment and macro-versus-micro averaging, so the two supplementary metrics answer different questions.

For paired method contrasts, we perform a paired PDF-cluster bootstrap over 49 source-report clusters. Each of 10,000 replicates samples 49 reports with replacement using seed 20260814, applies the same sampled rows to both methods and all three seeds, computes the within-seed differences, and averages those differences. The five pre-specified contrasts are M1–M0, M3–M2, M4–M1, M6–M3, and M6–M5. They form a separate Holm family of five for each metric, rather than one family of 20 [5]. Bracketed 95% intervals are uncorrected bootstrap percentile intervals; inferential decisions use $p_{\text{Holm}} < .05$, not whether an interval excludes zero.

6 RESULTS

Controlled decision comparison. Table 2 compares decision rules on identical base probabilities and Test rows. Independent argmax (M0) produces invalid tuples for 12.6% of document-disjoint predictions; projection and 17-state decoding (M1–M6) reduce this rate to zero by construction. The ± values are sample standard deviations across the three seeds. Because each seed determines both fold assignment and training randomness, they describe spread over the whole split-and-training pipeline, not a confidence interval.

Evaluation targets and robustness. Table 3 compares the same-document and document-disjoint targets. Its gaps, 0.012–0.015, separate estimands rather than estimate bias.

Table 3: Comparison of the two evaluation estimands. Same-document (seen-report, unseen-paragraph) versus document-disjoint (unseen-report) evaluation. The former matches the competition distribution because development and test share the 49 reports. Δ is a gap between two estimation targets, not a bias estimate; "Best calibrated projection" is M2 selected over M3, and "Best valid-state decoder" is M5 selected over M6, by document-disjoint mean official wF1 (the same values as in Table 2). Bracketed 95% report-bootstrap intervals are descriptive and are not adjusted for this outcome selection.

Method	Same-document	Document-disjoint	Δ	95% CI
M0	0.585	0.572	0.012	[0.004, 0.022]
Best calibrated projection	0.587	0.574	0.012	[0.001, 0.024]
Best valid-state decoder	0.590	0.576	0.015	[0.004, 0.027]

The C-wF1 result does not replicate under same-document evaluation: M1–M0 remains positive at +0.002 $[-0.001, 0.005]$ but does not survive Holm correction ($p_{\text{Holm}} = .739$).

Metric–constraint mismatch. Table 4 reports all 20 paired contrasts. For the pre-specified official-metric M1–M0 contrast, $\Delta = -0.001$ and the interval is $[-0.006, 0.003]$. With $p_{\text{Holm}} = 1.000$, there is no detectable difference. Post-hoc C-wF1 changes only whether an ancestor-unsupported child prediction receives credit. It gives +0.004 $[0.001, 0.007]$ and survives correction with $p_{\text{Holm}} = .025$. Post-hoc hF gives $\Delta = +0.003$ and corroborates the direction with $p_{\text{Holm}} = .017$; its interval is $[0.001, 0.005]$. However, hF also replaces macro with micro aggregation and therefore does not isolate consistency. Pre-specified tuple accuracy corroborates the direction with +0.035 $[0.028, 0.043]$ and $p_{\text{Holm}} = .001$.

Table 4: All pre-specified contrasts across four metric families. The five contrasts are evaluated on the document-disjoint protocol. A paired report-cluster bootstrap uses 10,000 resamples of source reports, not paragraphs; each metric is a separate Holm family of five. Bracketed 95% percentile intervals are uncorrected and do not determine significance; bold p_{Holm} marks survival of the correction. The official metric and tuple accuracy were pre-specified; C-wF1 and hF were adopted post hoc.

Metric	Contrast	Δ	95% CI	p_{Holm}
Weighted macro-F1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	-0.001	[-0.006, 0.004]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.007	[-0.017, 0.003]	0.643
	M6-M5	-0.009	[-0.017, -0.001]	0.135
Path-constrained wF1	M1-M0	+0.004	[0.001, 0.007]	**0.025**
	M3-M2	-0.001	[-0.006, 0.004]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.007	[-0.017, 0.003]	0.482
	M6-M5	-0.009	[-0.017, -0.001]	0.108
Hierarchical F1 (hF)	M1-M0	+0.003	[0.001, 0.005]	**0.017**
	M3-M2	+0.002	[-0.001, 0.005]	0.302
	M4-M1	+0.004	[0.000, 0.007]	0.135
	M6-M3	+0.006	[0.000, 0.012]	0.151
	M6-M5	+0.004	[-0.001, 0.009]	0.302
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.002	[-0.003, 0.007]	0.973
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.014, 0.004]	0.973
	M6-M5	-0.004	[-0.012, 0.004]	0.973

The complete family also contains adverse evidence. Compared with projection, uncalibrated 17-state decoding (M4–M1) lowers tuple accuracy by -0.006 $[-0.010, -0.002]$, $p_{\mathrm{Holm}} = .028$. Thus validity-preserving decoding is not uniformly superior. Searching every legal state need not improve whole-row correctness.

7 DISCUSSION

Within this controlled design, official wF1 and C-wF1 differ only in credit for an ancestor-unsupported child. Official M1–M0 is unresolved ($p_{\mathrm{Holm}} = 1.000$), whereas post-hoc C-wF1 survives correction ($p_{\mathrm{Holm}} = .025$). This localizes the mismatch to that scoring choice for this task, not a universal defect of field-wise metrics. Projection trades gains when the hierarchy forces *N/A* against replacing otherwise correct substantive children when an ancestor is wrong. Field-wise macro averaging and whole-row scoring value that exchange differently.

Metric choice also changes the point-estimate ranking. Official wF1 ranks M5 first, whereas hF ranks M6 first, so "best" is meaningful only together with the evaluation metric. Nor does a larger structured search guarantee a better tuple. A 17-state decoder can improve a parent decision without repairing the remaining children; conversely, changing a correct parent to an incorrect one necessarily destroys whole-row correctness. This asymmetry explains how a parent can improve while tuple accuracy declines, without implying that either validity-preserving rule is universally superior.

7.1 Limitations

The report-cluster bootstrap has an effective sample size of 49, limiting its resolution. Evidence comes from one dataset, language, ESG-report domain, and backbone. A seed jointly controls fold assignment and training randomness, so their effects cannot be separated. The *Misleading* class has only $n = 2$; we make no class-level improvement or significance claim. Biases optimize per-field macro-F1 before projection or decoding, so downstream legalization can undo a field-level gain. The calibrated arms may therefore understate the potential of decoder-specific metric-aware calibration.

We tested decision-time rules, not a structural training loss, direct 17-class classifier, classifier chain, LLM, or alternate backbone. We did not evaluate temporal or industry splits; the released rows contain no explicit report-date or industry field. The primary C-wF1 contrast is resolved only under document-disjoint evaluation ($p_{\mathrm{Holm}} = .025$), not same-document evaluation ($p_{\mathrm{Holm}} = .739$). Both C-wF1 and hF were selected post hoc. Training and decision hyperparameters use one frozen configuration without sensitivity analysis. Finally, defects in training-time gradient accumulation and in the original statistical decision rule were corrected before this analysis; all reported artifacts were regenerated after the corrections.

8 CONCLUSION

On this ESG dataset and backbone, field-wise scoring can fail to reflect validity-preserving decisions.

REFERENCES

[1] AI CUP n.d. *AI CUP 2026 競賽提交格式說明 Sample Submission Format Guide*. AI CUP. Official guide; no byline, version number, or issue date is stated in the document.

[2] Chung-Chi Chen, Yohei Seki, Hakusen Shu, Anaïs Lhuissier, Juyeon Kang, Hanwool Lee, Min-Yuh Day, and Hiroya Takamura. 2025. SemEval-2025 Task 6: Multinational, Multilingual, Multi-Industry Promise Verification. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Association for Computational Linguistics, Vienna, Austria, 2461–2471. https://aclanthology.org/2025.semeval-1.321/

[3] Yiming Cui, Wanxiang Che, Ting Liu, Bing Qin, and Ziqing Yang. 2021. Pre-Training With Whole Word Masking for Chinese BERT. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021), 3504–3514. https://doi.org/10.1109/TASLP.2021.3124365

[4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. https://doi.org/10.18653/v1/N19-1423

[5] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. https://www.jstor.org/stable/4615733

[6] Ke Ji, Yixin Lian, Jingsheng Gao, and Baoyuan Wang. 2023. Hierarchical Verbalizer for Few-Shot Hierarchical Text Classification. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Toronto, Canada, 2918–2933. https://doi.org/10.18653/v1/2023.acl-long.164

[7] Roman Plaud, Matthieu Labeau, Antoine Saillenfest, and Thomas Bonald. 2024. Revisiting Hierarchical Text Classification: Inference and Metrics. In *Proceedings of the 28th Conference on Computational Natural Language Learning*. Association for Computational Linguistics, Miami, FL, USA, 231–242. https://doi.org/10.18653/v1/2024.conll-1.18

[8] Tobias Schimanski, Julia Bingler, Mathias Kraus, Camilla Hyslop, and Markus Leippold. 2023. ClimateBERT-NetZero: Detecting and Assessing Net Zero and Reduction Targets. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 15745–15756. https://doi.org/10.18653/v1/2023.emnlp-main.975

[9] Chao Yu, Yi Shen, and Yue Mao. 2022. Constrained Sequence-to-Tree Generation for Hierarchical Text Classification. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Madrid, Spain). Association for Computing Machinery, New York, NY, USA, 1865–1869. https://doi.org/10.1145/3477495.3531765